Wearable Sensor For Multi-wavelength Near-infrared Spectroscopy Of Skin Hemodynamics Along With Underlying Muscle Electromyography

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GOAL

Monitoring of Intermittent claudication in peripheral arterial disease (PAD)

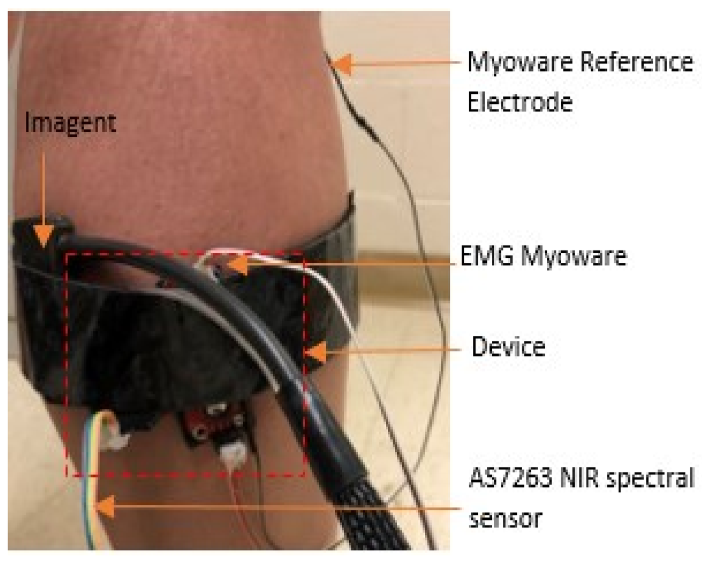
- Investigate skin near-infrared spectroscopy (NIRS) in conjunction with muscle activation using electromyography (EMG)
- Proof-of-concept wearable device:
  - Record: EMG, multi-wavelength NIRS (mwNIRS) spectra
  - Detect: muscle activation in conjunction with changes in skin blood volume

INTRODUCTION

- PAD occurs due to partial or complete obstruction of ≥1 peripheral arteries [1].
- More than 200 million people have PAD worldwide, where intermittent claudication is indicative of exercise-induced ischemic leg pain.
- Intermittent claudication also manifests in the skin perfusion [2]. In this single-subject study, we investigate the skin perfusion changes using NIRS during muscle activation measured with EMG.
- We developed a continuous wave skin mwNIRS-EMG sensor based on the principle of Beer-Lambert law to detect the variations in oxygenated (HbO) and deoxygenated (HHb) hemoglobin in the skin tissue in relation to muscle contractions (EMG).
- Skin mwNIRS sensor was based on a sensor configuration found using Monte Carlo-based simulations with a skin tissue model, that was compared with a state-of-the-art device frequency-domain NIRS system.

DEVICE

- **Continuous wave mwNIRS-EMG sensor**
- Arduino Uno Microcontroller (Arduino LLC, Italy)
- Myoware EMG sensor (Advancer Technologies LLC, USA)
- AS7263 NIR Spectral Sensor (SparkFun Electronics, USA)
- **Frequency domain (FD) NIRS** (Imagent, ISS, USA) – *muscle NIRS for comparison*



EXPERIMENTS

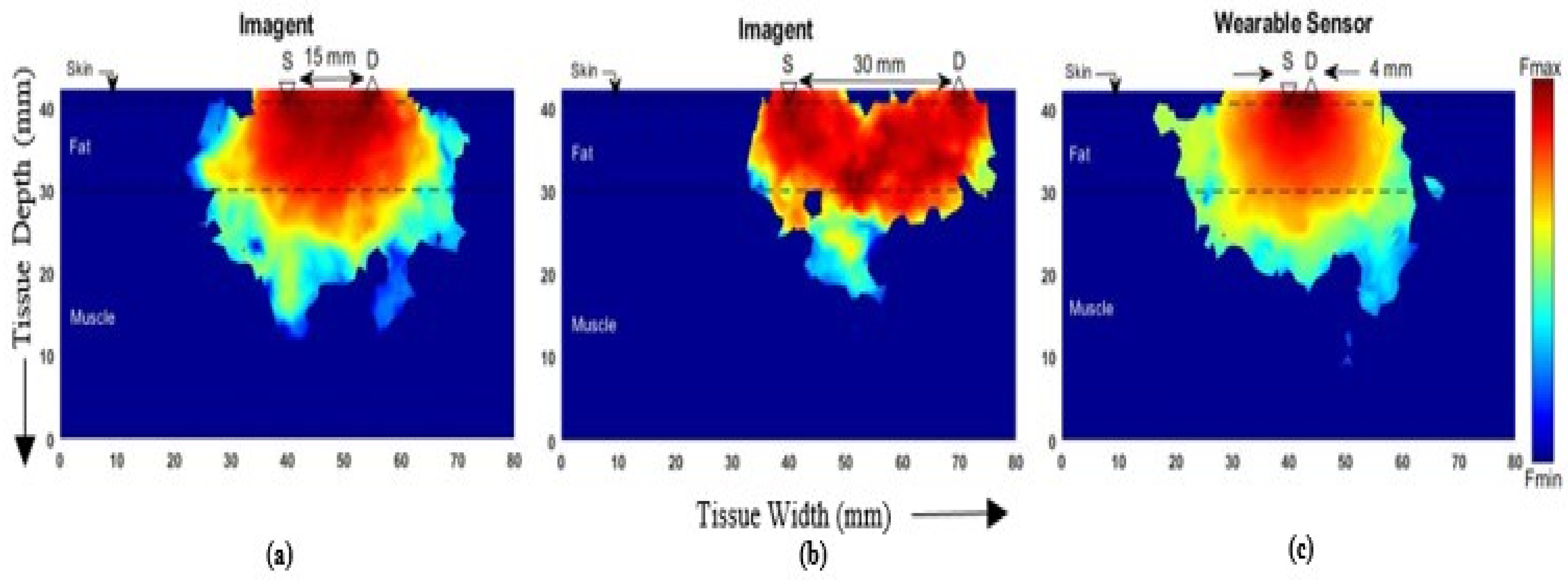
- A young and healthy subject (age: 25) volunteered for this case study.
- The experiment consisted of two sets of repetitions of calf raises at a frequency of 10 and 30 beats per minute (BPM), respectively.

SIGNAL PROCESSING

- The data from our skin mwNIRS-EMG sensor was analyzed using MATLAB R2018a.
- HbO and HHb concentrations in the skin blood were calculated using the Modified Beer-Lambert's equation with robust linear least squares regression using four (730nm, 760nm, 810nm, and 860nm) wavelengths.

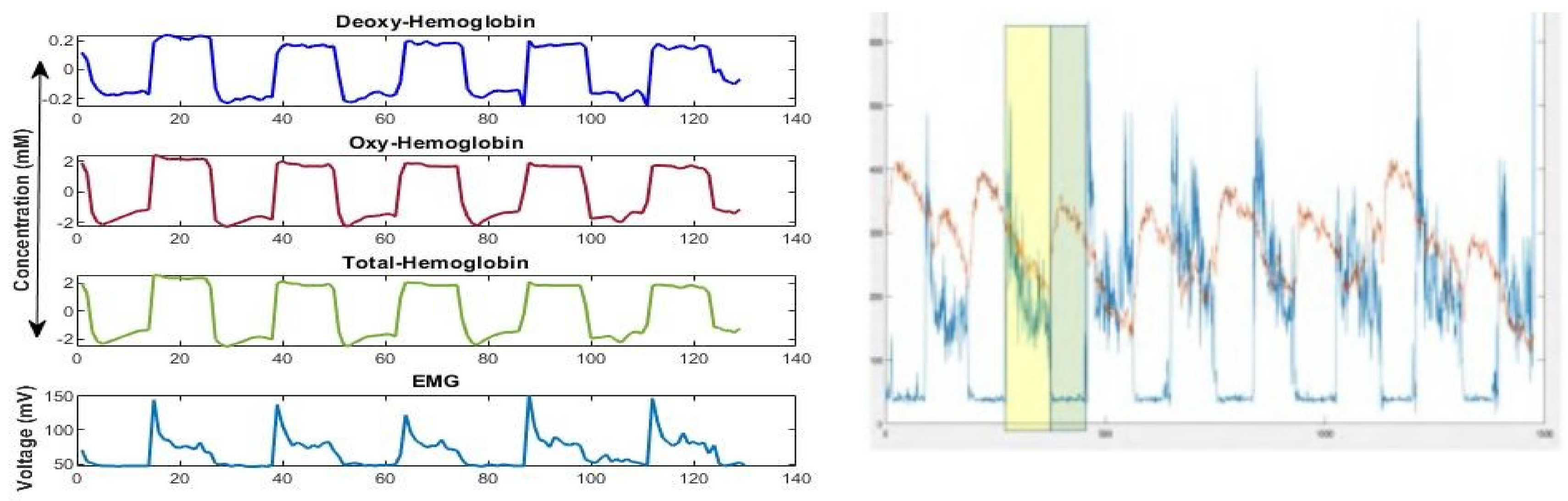
SIMULATION

- A simulation-based on the Monte Carlo model [3] was carried out to visualize the photon trajectory, from a virtual source of photons, through the calf tissues, i.e., skin, fat, and muscle.
- A 3-layer model representing the muscle, fat, and the skin tissues of thickness 30mm, 10.6mm [4], and 1.6mm [5] respectively was modeled using the MMCLAB (2017 edition).
- For FD-NIRS muscle sensor, a set of simulations were carried out for two wavelengths, i.e., 690nm and 830nm, at varying SD distances, i.e., 15mm, 20mm, 25mm, and 30mm.
- An isotropic photon source with an intensity of 10^6 incident photons was considered with simulation parameters, i.e., the refractive index (n), absorption coefficient (μa), reduced scatter coefficient (μs), and scattering anisotropy (g), for the three tissues taken from prior works [4,6].



- Photon trajectory and the variance in fluence rate within the three (skin, fat, muscle) tissue layers corresponding to the varying SD obtained using MC simulation; (a) FD-NIRS Imagent, SD = 15mm, (b) FD-NIRS Imagent, SD = 30mm and (c) Our wearable sensor, SD = 4mm.
- The high fluence rate of the photons is depicted by the red color tones and the low fluence rate with the blue shades
- Photons penetrated deeper (with higher fluence rate) into the muscle tissue with greater (>30 mm) SD, while their trajectory lies within the skin tissue for a lesser SD [7].

RESULTS



- The left panel of figure (above) shows the plots of <u>changes in HHb, HbO, and THb (blood volume) concentration in skin response to the muscle activation (rising curve) and deactivation (falling curve)</u> as indicated by the voltage plot from the linear envelope of the EMG
- The right panel shows FD-NIRS sensor data that showed <u>a delay in the muscle THb changes (green box) following muscle activation</u> (yellow box)

CONCLUSION

- The changes in the skin and muscle blood volume (THb) showed a relationship with the underlying muscle EMG in our single subject proof-of-concept study.
- Muscle NIRS provided important information, especially delay between THb and EMG, that can be separated from skin THb using a multi-distance continuous wave NIRS sensor in the future.

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